Global Climate Events Economic Impact

Data Dictionary | 2020 – 2025

2,929	20	50	6
Climate Events	Columns	Countries	Year Span

This document explains every column in the global climate events dataset. Each entry covers what the column means in plain language, its data type, valid range, and example values — so anyone can understand the data without prior technical knowledge.

COLUMN TYPE GUIDE

ID Date Text Number Geography Calculated

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1 Identification & Timing

`event_id` — ID

A unique code that identifies each climate event. No two rows share the same ID.

Format	Text code starting with "EV" + 5 digits
Example	EV01539, EV02303

`year` — Number

The year in which the event occurred. Useful for year-over-year trend analysis.

Range	2020 – 2025
Example	2022

`date` — Date

The exact calendar date when the event began.

Format	YYYY-MM-DD (year-month-day)
Example	2020-01-01

`month` — Number

The month the event started, stored as a number. Enables seasonal analysis (e.g. are floods more common in summer?).

Range	1 (January) – 12 (December)
Example	7 = July

2 Location

`country` — Text

The name of the country where the event took place.

Unique values	50 countries
Example	Japan, Canada, Brazil, Australia

`longitude` — Geography

The east-west position on the globe. Used together with latitude to place the event on a map. Positive = East, negative = West.

Range	-180 (far west) to +180 (far east)
Example	138.72 = central Japan

`latitude` — Geography

The north-south position on the globe. Combined with longitude, it pinpoints where the event happened on a map. Positive values = North, negative = South.

Range	-89.76 (far south) to +89.98 (far north)
Example	35.71 = northern Japan

3 Event Characteristics

`event_type` — Text

The category of climate or natural disaster that occurred.

12 possible values Cold Wave, Drought, Earthquake, Flood, Hailstorm, Heatwave, Hurricane, Landslide, Tornado, Tsunami, Volcanic Eruption, Wildfire

`severity` — Number

A score rating how destructive and dangerous the event was. Higher numbers indicate more severe events.

Scale	1 (minor) to 9 (catastrophic)
Example	3 = moderate impact, 8 = very severe

`duration_days` — Number

How many days the event lasted from start to end.

Unit	Days
Range	0 – 115 days
Example	6 = lasted 6 days

4 Human Impact

`affected_population` — Number

The total number of people whose daily lives were disrupted — including those evacuated, displaced, or cut off from basic services.

Unit	Number of people
Range	622 – 56,248,325 people
Example	420,956

`injuries` — Number

The number of people physically injured (but surviving) during the event.

Unit	Number of people
Range	0 – 734 injuries

`deaths` — Number

The number of people who died as a direct result of the event.

Unit	Number of people
Range	0 – 117 deaths

`total_casualties` — Calculated

Deaths and injuries combined into one figure, giving a quick snapshot of overall human harm caused by the event.

Formula	deaths + injuries
Range	0 – 738

5 Economic Impact

`economic_impact_million_usd` **Number**

The total financial damage caused by the event — covering destroyed property, lost businesses, and clean-up costs — measured in millions of US dollars.

Unit	Millions of USD (e.g. 5.0 = USD 5 million)
Range	USD 0 – USD 718.21 million
Median	USD 0.09 million (most events are small-scale)

`infrastructure_damage_score` **Number**

A composite score measuring how badly the event damaged physical infrastructure such as roads, bridges, buildings, and power lines. Higher means more destruction.

Range	0.2 – 63.0 (higher = worse)
Example	8.9 = significant damage

`impact_per_capita` **Calculated**

Economic damage divided by the number of people affected. Normalises the financial impact so that small localised events can be compared fairly with large-scale ones.

Formula	economic_impact_million_usd / affected_population
Range	USD 0 – USD 181.62 per person

6 Response & Aid

`response_time_hours` **Number**

How many hours elapsed before emergency responders arrived and began helping. A lower value means a faster, more effective response.

Unit	Hours
Range	0 – 60 hours
Example	11 = responders arrived within 11 hours

`international_aid_million_usd` **Number**

The amount of financial assistance received from other countries or international organisations to deal with the event.

Unit	Millions of USD
Range	USD 0 – USD 164.51 million
Note	Many events received USD 0 (handled domestically)

`aid_percentage` **Calculated**

The share of total economic damage that was covered by international aid. Expressed as a percentage.

Formula	international_aid / economic_impact x 100
Unit	Percentage (%)
Range	0% – 22.9%
Example	15 = 15% of damage was covered by external aid

Column type legend

ID	Unique identifier — one value per row
Date	Date or time value
Text	Category or free-text label
Number	Raw measured or counted value
Geography	Map coordinate (latitude or longitude)
Calculated	Derived from other columns in the dataset